Exam

Numerical Programming I
February 6, 2026

Name: ______________________________

Mixed Short Questions

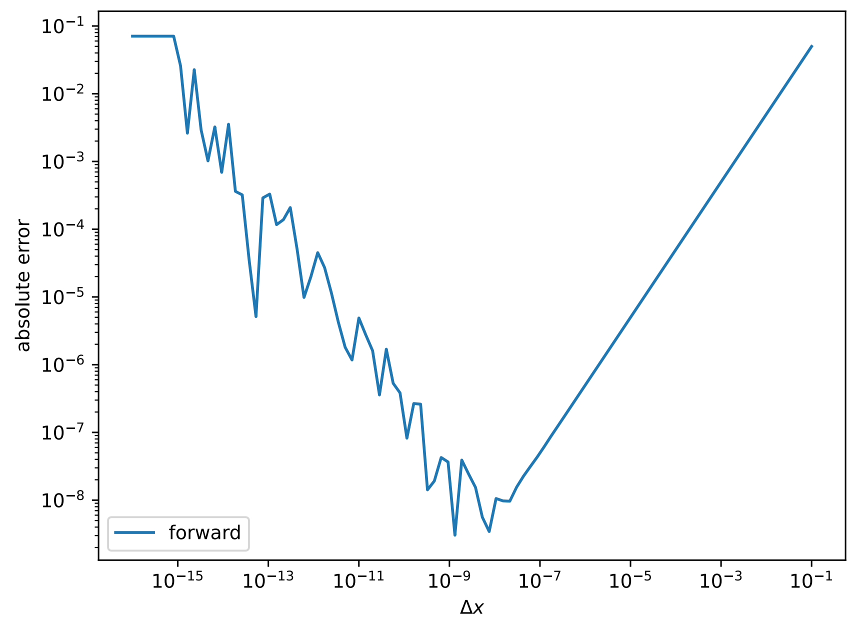
1. (a) What does $(1.0 + 10^{-17}) - 1.0$ evaluate to in double-precision floating point arithmetic (with machine precision $\epsilon_{\text{mach}} \sim 10^{-16}$)?

   ◯ 10^{-17} is correctly calculated.

   ◯ 10^{-17} cannot be represented in double-precision, the expression results in undefined behavior.

   ◯ The expression evaluates to zero, the mantissa of 1.0 does not have enough digits to capture 10^{-17}.

   (b) Consider a differentiable function $f : \mathbb{R} \to \mathbb{R}$. We have written a program which approximates $f'(x)$ using the forward difference $\frac{f(x+\Delta x)-f(x)}{\Delta x}$. Below, the absolute error $\left| f'(x) - \frac{f(x+\Delta x)-f(x)}{\Delta x} \right|$ over Δx is plotted. Mark the region where the floating point error is dominant and the region where the truncation error is dominant.



   (c) Does gradient descent always converge to the **global** minimum? Answer with yes or no and give a short justification.

Second order derivative approximation

2. For a twice differentiable function $f : \mathbb{R} \to \mathbb{R}$ we would like to find a finite difference approximation for the second derivative $f''(x)$.

   (a) Approximate $f(x + \Delta x)$ and $f(x - \Delta x)$ by Taylor expansion up to (including) third order around x.

   (b) Using these approximations find an approximation for $f''(x)$ based on $f(x)$, $f(x + \Delta x)$ and $f(x - \Delta x)$. What is the order of the error term? *Hint:* You need to combine both Taylor expansions such that the first and third order derivatives cancel.

   (c) For $f(x) = x^2$ and $\Delta x = 0.1$ approximate $f''(0)$ and compare your approximation to the analytical value.

Convergence of Newton's root finding method

3. In the lecture we have stated:

> If g is continuously differentiable in a neighborhood around a root $x^\star$ with nonzero derivative at $x^\star$, then there exists a neighborhood around $x^\star$ where Newton's iteration converges to $x^\star$ for all starting values.

Now consider the function

$$g(x) = x(1+x^2)^{-\frac{1}{2}} \tag{1}$$

(a) What is the root $x^\star$ of g where $g(x^\star) = 0$?

(b) Show that the derivative of g is $g'(x) = (1+x^2)^{-\frac{3}{2}}$. *Hint:* Calculate the derivative by product rule and factor out $(1+x^2)^{-\frac{3}{2}}$.

(c) Calculate $g'(x^\star)$. Is there a neighborhood around $x^\star$ where Newton's iteration converges to $x^\star$ for all starting values inside that neighborhood?

(d) Show that the Newton step

$$x_{k+1} = x_k - \frac{g(x_k)}{g'(x_k)} \tag{2}$$

is given by $x_{k+1} = -x_k^3$.

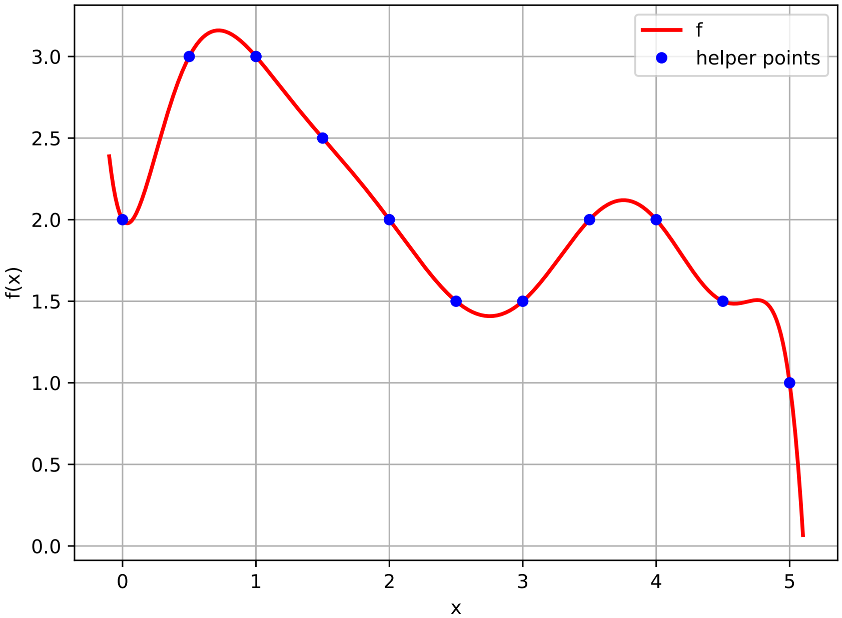
(e) Consider the following cases for the initialization of the Newton iteration

- $x_0 = 0.5$
- $x_0 = -1.0$
- $x_0 = 2.0$

and analyze if the iteration converges, diverges or oscillates.

(Bonus) Numerical integration by hand

4. Consider the function $f : \mathbb{R} \to \mathbb{R}$ plotted below.



We want to approximate the integral

$$\int_0^5 f(x)\,dx \tag{3}$$

Here, assume the subdivision

$$\int_0^5 f(x)\,dx = \sum_{i=0}^{N-1} \int_{x_i}^{x_{i+1}} f(x)\,dx \tag{4}$$

with $N = 5$ and $x_0 = 0.0, x_1 = 1.0, x_2 = 2.0, x_3 = 3.0, x_4 = 4.0, x_5 = 5.0$.

And consider the following *quadrature rules*:

- midpoint rule:

$$\int_{x_i}^{x_{i+1}} f(x)\,dx = \underbrace{(x_{i+1} - x_i)\, f\left(\frac{x_i + x_{i+1}}{2}\right)}_{=:M_i} + \mathcal{O}((x_{i+1} - x_i)^3) \tag{5}$$

- trapezoidal rule:

$$\int_{x_i}^{x_{i+1}} f(x)\,dx = \underbrace{(x_{i+1} - x_i)\left(\frac{f(x_i) + f(x_{i+1})}{2}\right)}_{=:T_i} + \mathcal{O}((x_{i+1} - x_i)^3) \tag{6}$$

(a) Approximate $\int_0^5 f(x)\,dx$ using the midpoint rule as given above.

(b) Approximate $\int_0^5 f(x)\,dx$ using the trapezoidal rule as given above.